

Try Your Hand with DevOps on Azure



The DevOps culture consists of continuous practices that help to achieve faster time to market and the highest quality for an application. This article gives a brief description of Azure DevOps, and then shows how to implement CI/CD in it for a sample application.

In agile software development, implementation of DevOps practices using end-to-end automation with Visual Studio Team Services or Azure DevOps helps to achieve better software quality and faster time to market.

Benefits of implementing DevOps practices

Table 1 lists the business and technical benefits realised by multiple organisations after the culture transformation achieved using DevOps practices.

Azure DevOps is a software as a service (SaaS), also

known as a hosted service. Visual Studio Team Services is a platform from Microsoft that provides an end-to-end automation toolchain for developing and deploying different types of applications written in Java, .NET, Python, PHP, Angular, Node.js, Ionic Cordova, Android, iOS, and so on. It integrates with most of the tools used in the automation of application life cycle management activities, and provides a free tier to get started. One of the major benefits of Azure DevOps is that it provides the hosted agents.

Table 2 gives the services available in Azure DevOps for implementing continuous practices.

Business benefits	Technical benefits
Improved communication and collaboration between teams	Early detection of failures
Happy people	High availability
Faster time to market	Stable and consistent environment
Quality releases	Efficient utilisation of resources
Productivity gains	Easy rollback: Enhanced recovery time
Transparency in process execution	Single-click deployment
Measurement of achievement; for example, improvement in deployment time	Continuous deployment
Agile teams	Monitoring
Reliability and reusability	Cross-skills learning

Table 1: DevOps benefits

CI/CD pipeline in Azure DevOps

Continuous integration (CI) is the DevOps practice of integrating new features or a bug fix in the existing distributed version control systems such as Git, which triggers code quality checks, unit test execution, and build. It is easy to implement, and hence for those starting with DevOps, it works as a solid base on which the rest of DevOps practices can stand. The objective of continuous delivery and continuous deployment (CI/CD) is to deploy an application in dev, test, UAT and the production environment in an automated manner. It helps in incremental releases after short spans of development, or a sprint in agile terms.

Some of the key terminologies used within Azure DevOps are:

- **Organisation:** We can create one or more organisations in Azure DevOps and each organisation can have multiple projects.
- **Project:** There are multiple projects in one organisation using access control. Each project has its own Azure Boards, Azure Repo, Pipelines, and other services

Let us now use the Azure DevOps Demo Generator to get a sample project and create a Pipeline.